An aerostructural reduced-order model

Identifying the LEI-V3 quasi-steady ROM from coupled simulations only

Model, identification method, uncertainty and validation

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Abstract

The quasi-steady reduced-order model (ROM) of Cayon, van Deursen and Schmehl [2] needs a lift and a drag polar, the bridle pitch θ_b between the bridle resultant and the wing chord, and a steering gain. This note identifies *every* one of them from coupled vortex-step-method (VSM) and Billow-wireframe solutions at the centre of the wind window. Near stall the coupled problem has more than one solution, and which one a continuation finds depends on the way it was reached: marching the angle of attack up stays attached far past the stall of the same wing marched down. The state marched down is the one that agrees with reality, so every state of the data set is moved to it and every polar is the march down. The lift and drag polynomials are selected by forward–backward stepwise regression on a cross-validation error that holds out whole frozen shapes, and include a smooth stall switch. The range of the controls the ROM may be flown in follows from the same data: a depower setting is valid if a gust does not stall it.

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1 Purpose and scope

The ROM is a point mass in the course reference frame whose wing is assumed to align instantaneously with the bridle resultant [2]. Its aerodynamic inputs are

- the wing lift and drag coefficients C_L, C_D, as functions of the wing angle of attack α_w and the controls;
- the geometric bridle pitch θ_b that turns the bridle angle of attack α_b into the wing angle of attack, $\alpha_w = \alpha_b - \theta_b$ (paper Eqs. 1–2);
- the steering gain $k_{\phi,s}$ of the control-induced aerodynamic roll, $\phi_{a,w} = k_{\phi,s}\, u_s$ (paper Eq. 43), which depends on the depower and on the steering itself.

The ROM (`rom_config_aerostructural.yaml`) is identified from simulations only: the VSM aerodynamics of the *deformed* kite, coupled to the Billow structural model.

The note is written so that the identification can be repeated with another aerostructural tool: Sect. 3 states the model, Sect. 4 defines every angle as a formula on vectors any tool provides, Sect. 5 builds the data set, Sect. 6 fits the ROM parameters to it and Sect. 7 sets the limits of the controls.

Scope and limits, stated once:

- centre of the wind window (azimuth and elevation 0), no reel speed, steady flight, gravity off (the ROM models gravity itself);
- the stall comes from VSM with artificial viscosity on frozen shapes;
- the coefficients describe the wing and its bridle lines: the ROM adds the KCU drag and the tether itself.

2 Overview

Two scripts do the identification, both reading the kite's `identification.yaml`: `examples/3_rom_identification/run_identification_sweep.py` solves the data set (Sect. 5) and `examples/3_rom_identification/identify_rom_aerostructural.py` fits the ROM and its limits (Sects. 6 and 7). Fig. 1 shows the steps. Flight data enters at none of them: it is compared with, and corrected against, afterwards.

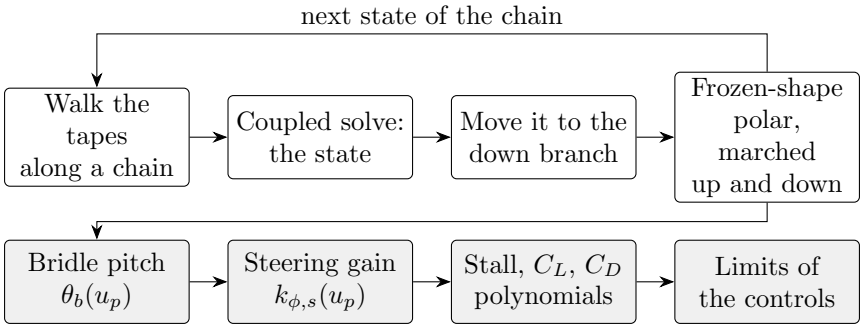


Figure 1: The identification. Top: the data set, solved per state. Bottom: the fit.

3 The model

3.1 Inputs and conventions

- u_p is the power-tape length l_{dp} in metres.
- u_s is the *standardised* steering of `awetrim.identification.controls`: $1.4\,\mathrm{m} \times u_s$ is the steering-tape half-difference (2019 KCU: $u_s = -\mathrm{kcu}/100$, full deflection ± 0.35); $u_s > 0$ turns the kite to its right.
- $\alpha \equiv \alpha_w$ is the angle of attack of the wing's centre-panel chord (Sect. 4).
- v_a is the apparent wind speed.
- All coefficients are referenced to $S = 19.75\,\mathrm{m}^2$, the EKF/flight area; the ROM file states it (`aerodynamics.reference_area`). It is *not* the projected area of the geometry ($17.21\,\mathrm{m}^2$).

A tool with another actuation model replaces u_p and u_s by its own inputs; what has to be identified against them is the same.

3.2 Lift and drag

Both coefficients are linear combinations of plain monomials,

$$C_L = c_0 + \sum_i c_i\, m_i(\alpha, u_p, u_s, v_a, \sigma), \qquad C_D = d_0 + \sum_j d_j\, m_j(\alpha, u_p, u_s, v_a, \sigma), \tag{1}$$

where $\sigma(\alpha)$ is a smooth attached-to-separated switch,

$$\sigma(\alpha) = \tfrac{1}{2}\left[1 + \tanh\left(\frac{\alpha - \alpha_s}{w}\right)\right], \tag{2}$$

with stall angle α_s and width w (ROM parameters `angle_of_attack_stall`, `width_stall`). Terms multiplied by σ act only past the stall, so $C = C_{\mathrm{att}} + \sigma\,(C_{\mathrm{sep}} - C_{\mathrm{att}})$ stays linear in its coefficients. Every term is smooth: the ROM evaluates monomials as written (a term may ask for $|m|$ with `abs: true`; the identification never does).

3.3 Bridle pitch and steering

$$\alpha_w = \alpha_b - \theta_b, \qquad \theta_b = \theta_0 + \theta_1\, u_p, \qquad \phi_{a,w} = k_{\phi,s}(u_p, u_s)\, u_s, \tag{3}$$

with θ_0, θ_1, $k_{\phi,s}$ the ROM parameters `angle_pitch_tether_0`, `slope_angle_pitch_tether_depower`, `gain_roll_steering`. This is the ROM's default steering law, `aerodynamics.steering_law: roll_gain`: the steering sets the wing's roll and the turn rate follows from the lateral force balance. The alternative, `steering_law: turn_rate`, is Sect. 3.4.

3.4 The turn-rate law

With `steering_law: turn_rate` the steering sets the *turn rate* instead, through the yaw-moment balance about the tether, and the roll is whatever tilt of the lift the lateral force balance needs. The law is Eq. (33) of Cayon and Schmehl [1], obtained by solving the lateral force, roll-moment and yaw-moment balances (their Eqs. 30–32) for the attitude and the relative turn rate $\dot{\chi}_{\text{turn}}$:

$$\dot{\chi}_{\text{turn}} = -\frac{k_1 v_a^2 u_s - mg\,(k_3 \sin\chi\cos\beta + k_5 \cos\chi\cos\beta + k_6 \sin\beta)}{k_4\, m\, v_\tau + k_2\, v_a}, \qquad k_3 = k_4, \tag{4}$$

with the coefficients of their App. A, Eq. (A3): k_1 the steering moment, k_2 the aerodynamic yaw damping, $k_3 = k_4$ the arm of the lateral load at the centre of gravity (weight and inertia act there alike), and k_5, k_6 the terms of the lateral centre-of-gravity offset x_n. $\dot{\chi}_{\text{turn}} = -a_n/v_\tau$ is the course rate without the transport of the course frame, m the kite mass and v_τ the tangential (ground) speed. Divided by k_2 and written with the weight vector $\mathbf{F}_g = -mg\,[\cos\chi\cos\beta,\ \sin\chi\cos\beta,\ \sin\beta]$ in the course frame (their Eq. 27), the ROM's residual is

$$\dot{\chi}_{\text{turn}}\,(v_a + b\,m\,v_\tau) + b\,F_{g,n} + c_5\,F_{g,\chi} + c_6\,F_{g,r} = a\,v_a^2\,u_s, \tag{5}$$

ROM parameter		unit	depends on
`gain_turn_steering` (+ `_depower`, `_squared`)	$a = -k_1/k_2$	$\text{rad}\,\text{m}^{-1}$ per u_s	u_p, u_s^2; not on ρ
`turn_inertia_coefficient` (+ `_depower`)	$b = k_3/k_2 = k_4/k_2$	kg^{-1}	u_p; $1/\rho$
`turn_gravity_course_coefficient`	$c_5 = k_5/k_2$	kg^{-1}	x_n; $1/\rho$
`turn_gravity_radial_coefficient`	$c_6 = k_6/k_2$	kg^{-1}	x_n; $1/\rho$
`turn_density_reference`	ρ_{ref}	$\text{kg}\,\text{m}^{-3}$	

$k_2 = \frac{1}{4}\rho S b^2(\dots)$ carries the density and k_3 to k_6 do not, so b, c_5 and c_6 are stated at ρ_{ref} and scale with ρ_{ref}/ρ. The total roll of the lift about v_a is then an unknown of the quasi-steady solve (`angle_roll`), closed by the lateral force balance; the turn law is a fourth residual row. The node NLP of the trajectory optimiser carries the roll as a per-node decision in the same way.

Why. In the roll-gain law the roll is the input, and the turn rate follows as $\dot{\chi} \approx L \sin\phi/(mv_\tau)$: it goes with the inverse of the kite's mass and of its *ground* speed. In the coupled model it does not. Solved without mass, the steered kite rolls by exactly zero, so the roll $\phi_{a,w}$ that the roll gain is identified on is entirely the tilt that supplies the centripetal force, and its turn rate per unit v_a is the same at every position in the wind window (within 0.6 %): the steering sets the turn through the yaw moment, as Eq. (4) states. At the centre of the window, where the identification sweep runs, v_τ/v_a is fixed per tape, so the two laws cannot be told apart there; they differ wherever the ground speed departs from the airspeed, that is off-centre and under gravity, which is everywhere in flight.

3.5 KCU drag, a separate force

C_D is the *wing's* drag, bridle lines included. The kite control unit is a separate bluff body hanging on the tether at the bridle point (ROM flag `kcu_drag_in_coefficients: false`), with the finite-cylinder law of `awetrim.aerodynamics.kcu_drag`,

$$\mathbf{F}_{\text{KCU}} = \tfrac{1}{2}\rho\left(A_\parallel\,|\mathbf{v}_\parallel|\,\mathbf{v}_\parallel + A_\perp\,|\mathbf{v}_\perp|\,\mathbf{v}_\perp\right), \tag{6}$$

where $\mathbf{v}_\parallel$, $\mathbf{v}_\perp$ split the apparent wind about the tether, and the drag areas come from the system file's KCU length, diameter and onboard turbine. The KCU drag enters the total aerodynamic force *and* the bridle resultant $\mathbf{F}_b$ (tether plus the KCU's weight, inertia and drag), so swapping the KCU changes $\mathbf{F}_b$ without re-identifying the wing.

4 Geometric definitions

Every angle the ROM needs is stated here as a formula on vectors, so that the parameters can be extracted from any tool's output without AWETrim. Fig. 2 draws them. Vectors are in any common Cartesian frame.

The two angles of attack. They have different roles:

- α_w, the angle of the apparent wind to the *centre-panel chord*, is the independent variable of the polars. Every C_L, C_D sample, from a trim or from an angle-of-attack sweep, is recorded against α_w, and the ROM evaluates its polynomials at α_w.
- α_b, the angle of the apparent wind to the *bridle resultant* $\mathbf{F}_b$, is what the point-mass ROM computes from its own forces. It serves one purpose: the bridle pitch $\theta_b = \alpha_b - \alpha_w$, the angle between the chord and the bridle resultant at trim, set by the bridle geometry and the depower setting, through which the ROM recovers $\alpha_w = \alpha_b - \theta_b$.

Symbols.

- $\mathbf{v}_a$: the apparent wind at the wing, the velocity of the air relative to the kite (it flows from leading to trailing edge); $\hat{\mathbf{v}}_a = \mathbf{v}_a/|\mathbf{v}_a|$.
- $\mathbf{e}_r$: the unit radial direction, ground station to kite.
- $\mathbf{F}_w$: the aerodynamic force on the wing *and its bridle lines*, without the KCU and the tether.
- $\mathbf{F}_b$: the bridle resultant, the force the bridle lines carry to the wing from the bridle point B. It is the tether force at B plus the KCU's weight, inertia ($-m_{\mathrm{KCU}}\mathbf{a}$) and drag; equivalently, from the wing's own balance with wing mass m_w, $\mathbf{F}_b = -\big(\mathbf{F}_w + m_w(\mathbf{g} - \mathbf{a})\big)$, which for a gravity-off trim without acceleration is $-\mathbf{F}_w$.
- P_{LE}, P_{TE}: the mid-span leading- and trailing-edge points of the *centre panel* (the panel spanning the centreline, between the two innermost struts) of the *deformed* wing; $\mathbf{s}$ that panel's span direction, from the left to the right strut.

Wing axes. Aircraft FRD axes anchored to the centre panel:

$$\begin{aligned} \mathbf{x}_K &= \frac{P_{\mathrm{LE}} - P_{\mathrm{TE}}}{|P_{\mathrm{LE}} - P_{\mathrm{TE}}|} && \text{chord, forward,} \\ \mathbf{z}_K &= \frac{\mathbf{x}_K \times \mathbf{s}}{|\mathbf{x}_K \times \mathbf{s}|} && \text{down, toward the bridle,} \\ \mathbf{y}_K &= \mathbf{z}_K \times \mathbf{x}_K && \text{right wing.} \end{aligned} \tag{7}$$

Wing angle of attack. With $\mathbf{u} = -\mathbf{v}_a$ the kite's velocity through the air,

$$\alpha_w = \operatorname{atan2}\big(\mathbf{u}\cdot\mathbf{z}_K,\ \mathbf{u}\cdot\mathbf{x}_K\big), \tag{8}$$

positive nose up.

Lift and drag coefficients. $D = \mathbf{F}_w \cdot \hat{\mathbf{v}}_a$, $L = \big|\mathbf{F}_w - D\,\hat{\mathbf{v}}_a\big|$ (positive on the $-\mathbf{z}_K$ side), $C_{L,D} = \{L, D\}/(\frac{1}{2}\rho v_a^2 S)$ with $S = 19.75\,\mathrm{m}^2$.

The apparent-wind frame. The ROM takes the bridle angles in the frame of paper App. D, built from $\mathbf{v}_a$ and $\mathbf{e}_r$ only:

$$\mathbf{e}_{r'} = \frac{\mathbf{e}_r - (\mathbf{e}_r \cdot \hat{\mathbf{v}}_a)\hat{\mathbf{v}}_a}{|\cdot|}\ \text{(zero-roll lift direction)}, \qquad \mathbf{e}_p = \frac{\mathbf{v}_a \times \mathbf{e}_r}{|\mathbf{v}_a \times \mathbf{e}_r|}\ \text{(toward the left wing)}. \tag{9}$$

($\mathbf{e}_p = -\mathbf{e}_{n'}$ of paper Eq. D6; in a straight flight $\mathbf{e}_p \approx -\mathbf{y}_K$.)

Bridle angle of attack and bridle pitch. With $\mathbf{P} = -\mathbf{F}_b$, the pull of the wing on the bridle,

$$\alpha_b = \operatorname{atan2}\big(\mathbf{P}\cdot\hat{\mathbf{v}}_a,\ \mathbf{P}\cdot\mathbf{e}_{r'}\big) \quad \text{(paper Eq. D11)}, \qquad \theta_b = \alpha_b - \alpha_w. \tag{10}$$

α_b is the angle by which the pull leans downstream from $\mathbf{e}_{r'}$; for a massless kite without KCU drag it is the glide angle, $\tan\alpha_b = C_D/C_L$, a useful check. In a symmetric trim ($\mathbf{v}_a$, $\mathbf{e}_r$ and $\mathbf{F}_b$ in the wing's symmetry plane) the bridle pitch follows from the trimmed geometry alone, independent of the inflow:

$$\sin\theta_b = \mathbf{x}_K\cdot\hat{\mathbf{F}}_b, \tag{11}$$

the angle by which the centre chord's leading edge dips toward the bridle resultant from the perpendicular. $\theta_b > 0$ (the LEI-V3 values) means the nose is pitched down relative to that perpendicular, so $\alpha_w < \alpha_b$. In the paper's terms $\theta_b = \theta_d + \lambda_b$ (Eq. 2); the ROM does not split it, it fits $\theta_b(u_p)$ directly. θ_b is always taken against the bridle resultant, never against the tether direction: the two differ by the KCU loads.

Roll angles. The roll of a force $\mathbf{F}$ about $\mathbf{v}_a$, zero along $\mathbf{e}_{r'}$ and positive toward the left wing, is

$$\phi(\mathbf{F}) = \operatorname{atan2}\big(\mathbf{F}\cdot\mathbf{e}_p,\ \mathbf{F}\cdot\mathbf{e}_{r'}\big). \tag{12}$$

The bridle roll is $\phi_{a,b} = \phi(\mathbf{P})$, the total aerodynamic roll $\phi_a = \phi(\mathbf{L})$ with $\mathbf{L} = \mathbf{F}_w - D\,\hat{\mathbf{v}}_a$, and the steering roll is their difference, $\phi_{a,w} = \phi_a - \phi_{a,b} = k_{\phi,s}u_s$ (paper Eqs. 22, D13–D14). A positive u_s turns the kite to the right as seen from the kite; with $k_{\phi,s} < 0$ it tilts the lift toward the right wing ($\phi_{a,w} < 0$) and the course rate is $\dot{\chi} > 0$. Note for readers of the paper: App. D1 describes a positive ϕ_a as a right-hand turn, but its Eqs. D6–D9 (and the code) tilt the lift toward the left wing for $\phi_a > 0$; this note follows the equations. Main-text Eq. 24 also has the opposite sign of the $\sin\phi_a$ terms to Eq. D9; the code uses Eq. D9.

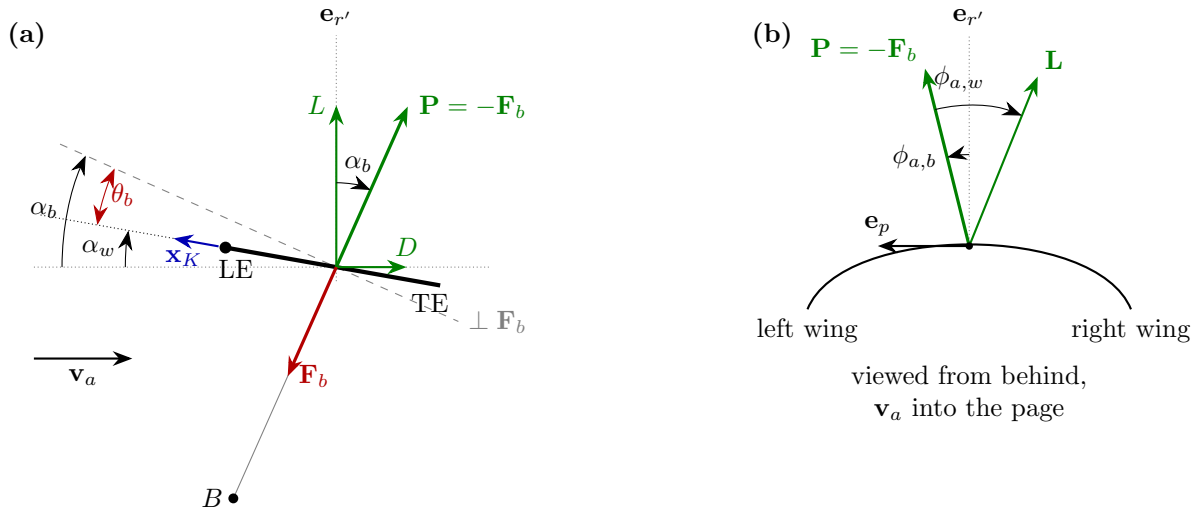


Figure 2: Geometric definitions (angles exaggerated). (a) Side view in the wing's symmetry plane: centre-panel chord with axis $\mathbf{x}_K$, apparent wind $\mathbf{v}_a$, the wing's lift and drag, the bridle resultant $\mathbf{F}_b$ acting on the wing from the bridle point B and its reverse $\mathbf{P}$ (collinear with $\mathbf{F}_w$ for a massless kite). α_w is the chord angle from the oncoming flow, α_b the angle of the line perpendicular to $\mathbf{F}_b$ (equally, of $\mathbf{P}$ from $\mathbf{e}_{r'}$), and $\theta_b = \alpha_b - \alpha_w$ the bridle pitch fixed by the bridle geometry and depower setting. (b) Rear view about $\mathbf{v}_a$: the bridle roll $\phi_{a,b}$ of $\mathbf{P}$ and the steering roll $\phi_{a,w}$ of the lift relative to it, both measured positive toward the left wing; drawn for $u_s > 0$ (right turn, $\phi_{a,w} < 0$).

5 The data set

Every state of the data set is reached along a chain, solved by the coupled model, moved to the realistic stall branch, and then given a polar. Two kinds of data come out, and the second is not optional: converged trims, and angle-of-attack sweeps on their frozen shapes.

5.1 The chains

All states are at the centre of the wind window: course 90°, no reel speed, steady flight, uniform wind, gravity off. A chain is a sequence of states, each starting from the one before. A depower chain ends when its stalled span fraction exceeds 0.2, a steering chain at the end of its range, and either at a state that does not converge.

- **Depower chains**, at zero steering, walk the power tape from the length the geometry is built at past both ends of the band, at several wind speeds, reaching the apparent wind speeds of reel-out and reel-in. They give $\theta_b(u_p)$ and the depower and load terms of the polars.
- **Steering chains** walk the power tape to a set length, then grow the steering half-difference (one side suffices; left and right are mirror images), at lengths across the band including its powered end. They give $k_{\phi,s}(u_p)$, the steering terms and the steering limit.

Table 1: The LEI-V3 chains, on the 2019 hardware.

Chain	Range	Wind speed
Depower	u_p from 1.90 m to 1.10 m, steps of 0.05 m	4, 5, 6 m s^{-1}
Depower	u_p from 1.90 m to 2.60 m, steps of 0.05 m	4, 5, 6, 7, 8 m s^{-1}
Steering	half-difference to 0.50 m, steps of 0.025 m, at u_p = 1.40, 1.50, 1.70, 1.95, 2.10, 2.40 m	4, 5, 6 m s^{-1}

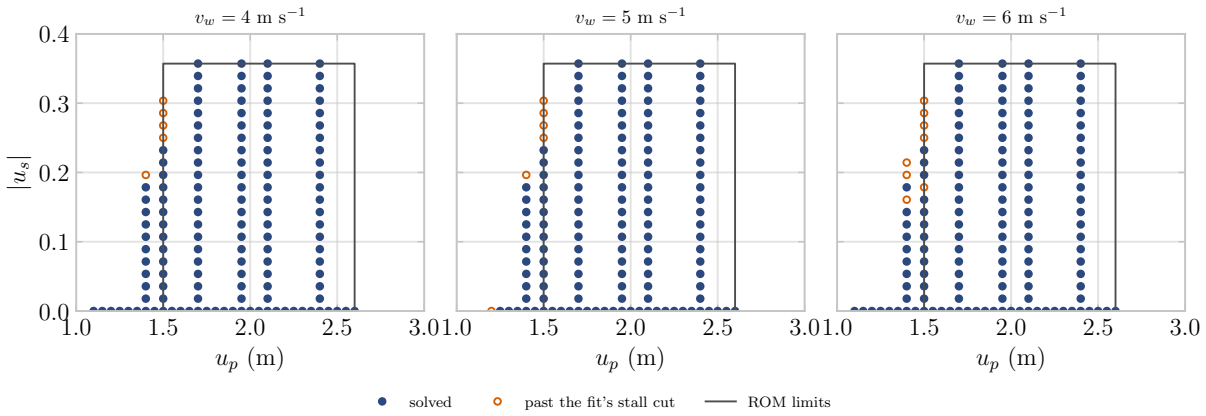


Figure 3: Every solved state in the plane of the controls, at the three wind speeds with steering chains (7 and 8 m s^{-1} have depower chains only). A chain ends at the first state past the stall stop (open markers) or that does not converge. Box: the ROM's limits.

5.2 The coupled solve

The kite's structural file is reduced to the wireframe (cables, tension-only lines, pulleys and the wing's struts), and the wing is meshed with 135 VSM panels on the billowed geometry, the loads transferred by the strut-lever rule. At each state the coupled loop moves the tapes, solves the structure by energy minimisation, trims the kite with a Newton solve on the trim states and the circulation, transfers the loads and repeats until the forces settle. The trim balances the forces and the moments about the bridle point, with the tether force at that point, and solves for the flight speed and, for a steered state, the course rate. A Newton solve that does not reach its tolerance reports the lowest residual it visited, and is accepted only near a root.

5.3 The stall branch

A wing past its stall has more than one solution for the circulation, and a seeded solve stays on the one it starts near. The chains approach the powered tapes from the attached side, so they stay attached past the point at which the same wing, marched down in angle of attack from a fully stalled state, is already separated (Fig. 4). The march down is the stall that agrees with reality, and it is the one identified:

1. from the converged state, freeze the shape and march the angle of attack up past the stall and back down to the trim's own inflow, keeping that circulation;
2. solve the coupled problem again from it, structure and trim free.

A state with one solution comes back to itself. A state with two stays on the lower one, at a higher angle of attack and a lower tension, and that re-solved state is the data point. Step 2 is the test: the frozen march alone does not say that the kite stays there, and the coupled solve is free to go back. The maximum of the march must be well past the stall, or the march down has not fully separated and its stall reads too high.

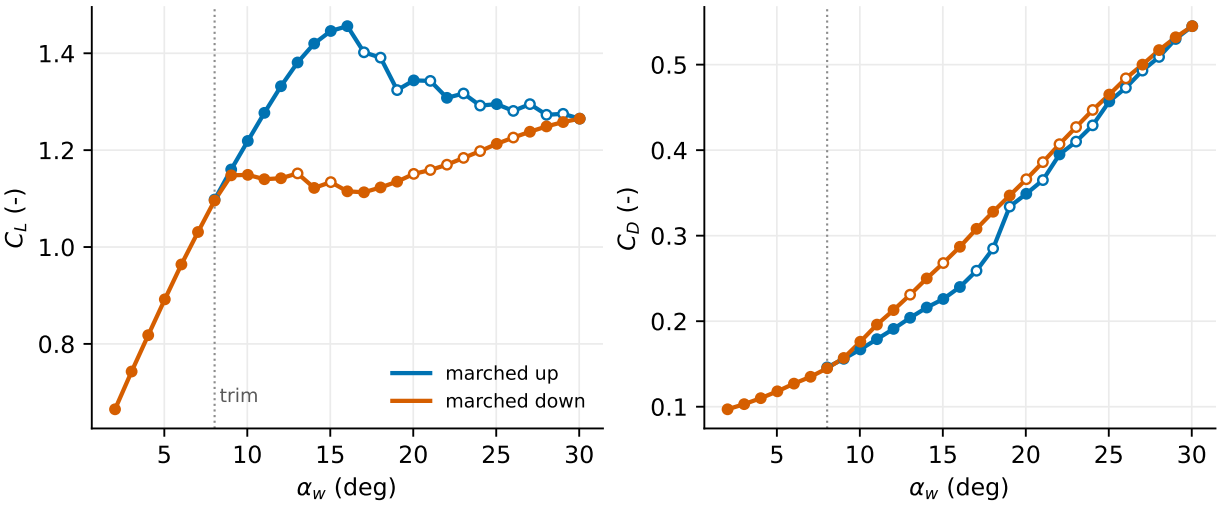


Figure 4: The two branches of one frozen shape: LEI-V3, $u_p = 1.40\,\mathrm{m}$, no steering, $5\,\mathrm{m\,s^{-1}}$ wind. Coefficients on $S = 19.75\,\mathrm{m}^2$ (multiply by 1.147 for the projected $17.21\,\mathrm{m}^2$). Open markers: the solve did not converge. The march down stalls near 10°, the march up near 16°.

5.4 The polars

Trims alone cannot identify the polars. Along a depower chain α_w follows u_p almost one to one, so a fit on trims cannot tell the polar slope from the depower effect, while in flight gravity and turning move α_w by several degrees around the trim value. Around every state:

1. freeze the converged deformed shape, and keep $|\mathbf{v}_a|$, the sideslip and the body rates of the trim;
2. solve the aerodynamics at $\Delta\alpha_w = 0$ from the trim's own circulation, which returns the trim's force;
3. rotate the inflow about $\mathbf{y}_K$ in steps of 1°, up to 22° about the trim, and then back down to −6°, each solve seeded with the one before and without re-trimming (the moment about the bridle point is not balanced; this is the polar of that shape);
4. record α_w (Eq. 8), $\mathbf{v}_a$, $\mathbf{F}_w$ and, separately, $\mathbf{F}_{\mathrm{KCU}}$ at each step of the march *down*.

The $\Delta\alpha_w = 0$ step must reproduce the trim's force within 0.1 %, or the sweep is discarded. Solves that do not converge, which happen above the stall, are dropped from the fit.

5.5 What each state must output

With these, another aerostructural tool can repeat the identification.

Quantity	Definition
u_p, u_s	controls of the trim
ρ, wind speed	air density, free-stream wind
$\mathbf{v}_a$	apparent wind at the wing reference point (air relative to kite)
$\mathbf{e}_r$	radial unit vector, ground station to kite
$\mathbf{F}_w$	aerodynamic force on wing and bridle lines, without KCU
$\mathbf{F}_{\mathrm{KCU}}$	KCU aerodynamic drag, separately
$\mathbf{F}_t$	tether force at the bridle point
m_w, m_{KCU}, $\mathbf{a}$	wing and KCU mass, kite acceleration (or the inertial force), to build $\mathbf{F}_b$
P_{LE}, P_{TE}, $\mathbf{s}$	centre-panel chord points and span direction of the *deformed* wing
v_τ, $\dot{\chi}$	trimmed tangential speed and course rate
deformed shape	the full converged structural geometry, to be frozen for the polars
convergence	flag and force/moment residuals

From these, Sect. 4 gives per trim α_w, C_L, C_D, α_b, θ_b, $\phi_{a,b}$ and $\phi_{a,w}$.

6 The fit

The samples are the converged states inside the fit band of the power tape (Sect. 7), and the polars that pass the check of Sect. 5.4.

6.1 Bridle pitch

$\theta_b = \alpha_b - \alpha_w$ (Eq. 10) is fitted linearly in u_p by ordinary least squares on the unsteered trims. With gravity off the coupled equilibrium is $\mathbf{F}_w + \mathbf{F}_{\mathrm{KCU}} + \mathbf{F}_t + \mathbf{F}_{\mathrm{in}} = 0$, with $\mathbf{F}_{\mathrm{in}}$ the inertial force, and the bridle resultant is $\mathbf{F}_b = -(\mathbf{F}_w + \frac{m_w}{m}\mathbf{F}_{\mathrm{in}})$.

6.2 Steering gain

$\phi_{a,w}$ per u_s (Sect. 4) is fitted as a polynomial in the depower plus an even term in the steering, $k_{\phi,s}(u_p, u_s) = k_0 + k_1 u_p + k_2 u_p^2 + (s_0 + s_1 u_p)\, u_s^2$, with both degrees chosen by leave-one-chain-out cross-validation, on the steered trims with a stalled span fraction up to 0.2.

6.3 Turn-rate law

For `steering_law: turn_rate`, Eq. (5) has two sets of unknowns that the centre-of-window sweep does not separate: with the mass fixed and v_τ/v_a fixed per tape, it identifies only $a/(1+b\,m\,v_\tau/v_a)$, as the paper notes for the flight data ("nearly collinear regressors"). The coefficients are therefore identified in two steps, both on the sweep's own states.

1. The sweep solves a few extra states from its anchors, tetherless (`turn_law:` in the sweep block, `run_identification_sweep.py --turn-law`, written to `sweep/turn_law/`). On one steered anchor per tape, the centre of the window at mass scale 1 and 0, gravity off: without mass the law reduces to $\dot{\chi}_{\mathrm{turn}} = a\, v_a u_s$, which gives a; the full-mass state then gives b, and $b(u_p)$ is a polynomial over the tapes (`fit.turn_law.inertia_degree`, linear for the LEI-V3). On one anchor, gravity-on states at courses 0 to 270° and elevations 10 to 60°: what the k_3 term leaves, by least squares, gives c_5 and c_6. ρ_{ref} is the states' density (`awetrim.identification.turn_law`).
2. $a(u_p, u_s)$ on all steered trims of the sweep with those terms, with the roll gain's form and its cross-validated degrees: `identify_rom_aerostructural.py --steering-law turn_rate` (`--steering-only BASE_ROM` refits the steering law alone and keeps the rest of a ROM).

The measurement is part of the sweep so that b always belongs to the shapes a is fitted on. `check_window_position.py` re-solves anchors at other positions in the window, masses and gravity, and checks the identified laws there.

6.4 Stall location

α_s is measured, not fitted: it is the median stall onset (Sect. 7) of the unsteered states near their stall, $u_p = 1.40$ to $1.70\,\mathrm{m}$ for the LEI-V3, which gives $10.27°$, with $w = 1°$. Chosen on the fit instead, it would follow the many samples of the plateau rather than the few at the lift drop. The coefficients are then fitted on the samples up to $8°$ past α_s, on states with a stalled span fraction up to 0.2. The ROM is not flown past the stall, so its operating range is what the fit serves; past α_s it follows the trend of the data, not the lift drop itself.

6.5 Candidate terms

Group	Terms	Reason
attached polar	α, α^2	lift slope and curvature
depower	u_p, u_p^2, αu_p, $\alpha^2 u_p$	shape change with the power tape
steering	u_s^2, αu_s^2, $u_p u_s^2$	symmetric in left/right steering
load	v_a, αv_a, $u_p v_a$	deformation grows with dynamic pressure
stall	σ, $\sigma\alpha$, $\sigma\alpha^2$, σu_p, σu_s^2	active past the stall only

There is no α^3: the ROM's solvers do not bound α, a cubic C_D went negative beyond the data, and a cubic C_L cannot follow the sharp stall.

6.6 Least squares and selection

For a term set $\mathcal{T}$, the coefficients minimise the sum of squared residuals, $\hat{\mathbf{c}} = \arg\min_{\mathbf{c}} \|\mathbf{y} - X_{\mathcal{T}}\mathbf{c}\|_2^2$, solved with each column scaled to unit RMS for conditioning and mapped back.

Polar samples of one frozen shape are strongly correlated, so the selection score is the *grouped* K-fold cross-validation error with whole anchors held out ($K = 5$):

$$\mathrm{CV}(\mathcal{T}) = \frac{1}{K}\sum_{k=1}^{K}\sqrt{\frac{1}{|G_k|}\sum_{i\in G_k}\left(y_i - \hat{y}_i^{(-k)}\right)^2}, \tag{13}$$

where $\hat{y}^{(-k)}$ is fitted without fold G_k. A random split would score interpolation inside a polar rather than prediction at an unseen state. BIC is not used: it treats the correlated samples as independent evidence and keeps accepting terms worth a fraction of a percent.

1. **Forward**: from the intercept, add the candidate that lowers CV most; stop when the best improves it by less than 1 %.
2. **Backward**: drop any term whose removal costs less than the same 1 %; a term picked early can become redundant later.
3. **Extrapolation check**: C_D should stay positive over $\pm 30°$ across the fitted u_p, u_s, v_a ranges; the script warns if it does not.

7 The limits of the controls

The ROM is valid in a box of the controls, written to its `validity` block, to which the optimiser's bounds are clipped.

Depower. A depower setting that is itself stalled is no operating point, and one a gust can stall is not a safe one. The powered end of the band is the shortest tape for which a gust stays below the stall onset of the march down:

$$\alpha_w + \arctan\frac{\Delta v}{v_a} < \alpha_{\text{onset}}, \tag{14}$$

with $\Delta v = 1\,\mathrm{m\,s^{-1}}$ and α_{onset} the angle at which C_L falls 5 % below the attached curve extrapolated from the polar below the trim. For the LEI-V3 this gives 1.50 m. The fit band reaches further, to the shortest tape not itself stalled, 1.40 m: no stalled depower setting enters the fit, its polars still reach past the stall, and the edge of the ROM's band is not an extrapolation. The depowered end is the end of the sweep, 2.60 m. Figures 5 and 6 show this along the depower chains.

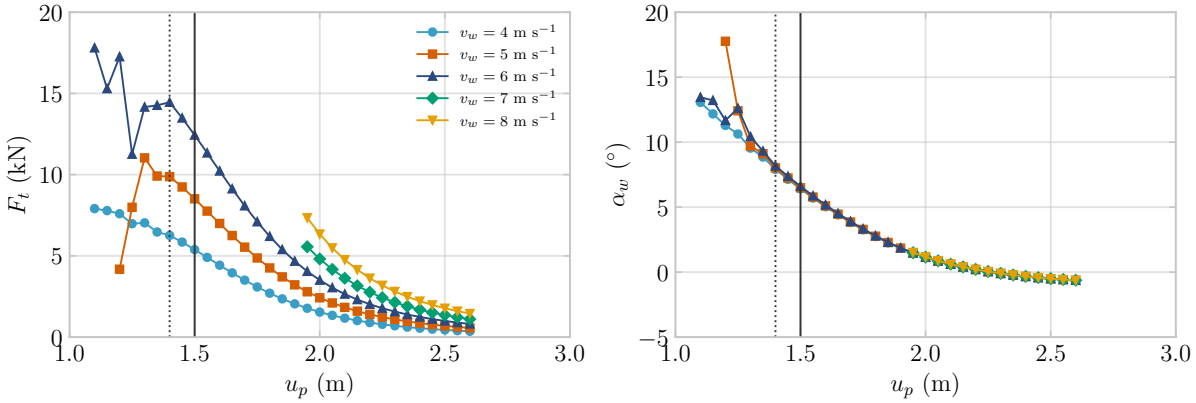


Figure 5: Tension and angle of attack along the depower chains, on the realistic stall branch. Solid line: the ROM's powered end; dotted line: the end of the fit band. Below the dotted line the states stall and the curves stop being smooth.

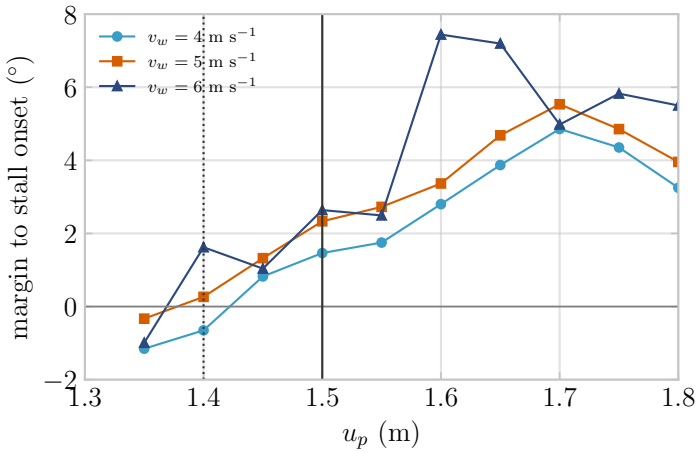


Figure 6: Margin to the stall onset after a gust of $1\,\mathrm{m\,s^{-1}}$, $\alpha_{\text{onset}} - \alpha_w - \arctan(\Delta v/v_a)$. The onset is read against a curve extrapolated from below the trim, so the measure holds only for states near their stall, which is the range shown.

Angle of attack. The optimiser's upper bound is the stall less a margin for the same gust, 3° for the LEI-V3, so 7.27°: the ROM's `validity` states what may be flown, not how far the data reach.

Steering. Steering stalls one wing tip, earlier the more powered the kite is (Fig. 7). Steering chains therefore do not stop on stall: they run to the end of their range or to a state that does not converge, which is what shows the limit. At 1.50 m they end at a failed trim at 0.45 m, with the tension falling in steps from about 0.10 m, while the longer tapes reach the flown range. The course rate keeps rising almost to the end, so it does not define a limit. The limit itself is still open (Sect. 10).

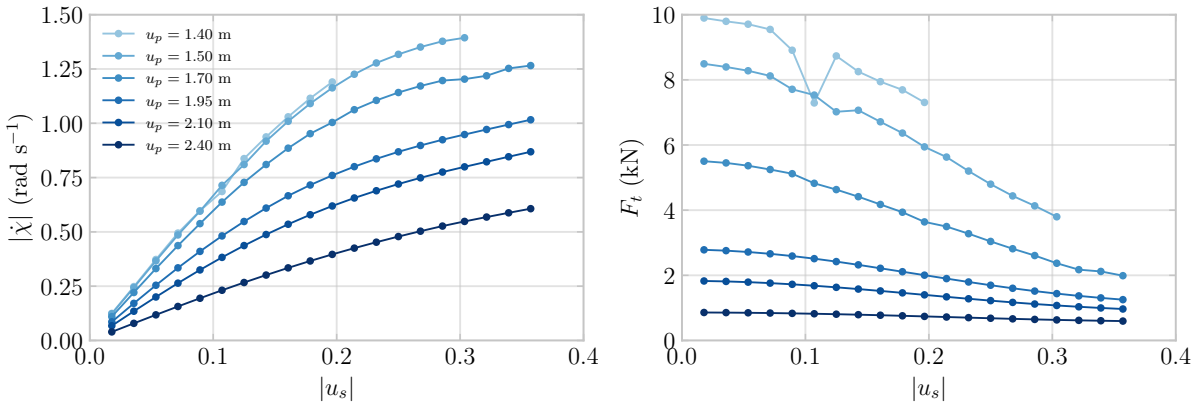


Figure 7: Course rate and tension against steering along the steering chains, 5 m s^{-1} wind.

Tension. The force margin depends on the wind, so it is a tension floor for the optimiser rather than a limit on the tape: the tether force must stay above twice the KCU weight after a 15 % lull in speed (602 N for the LEI-V3).

8 Results

The LEI-V3 data set has 444 coupled states. The fit uses 410 of them, with 7071 samples.

8.1 Identified coefficients

Table 2: Selected C_L and C_D terms (α in rad, u_p in m, σ the stall switch of Eq. 2), on both reference areas.

Coefficient	Term	On 19.75 m^2	On 17.21 m^2
C_L	1	+0.5978	+0.6859
	α	+4.4177	+5.0691
	$\sigma\,\alpha^2$	−4.1966	−4.8154
	$\alpha\,u_s^2$	−6.2396	−7.1597
	u_s^2	−4.8487	−5.5637
	α^2	−6.9232	−7.9441
	$\sigma\,u_p$	+0.0366	+0.0420
	u_p^2	−0.0334	−0.0383
	$u_p\,u_s^2$	+1.8198	+2.0881
	$\alpha^2\,u_p$	+1.9091	+2.1906
C_D	1	+0.0892	+0.1024
	α^2	+1.6202	+1.8591
	$u_p\,u_s^2$	−0.0667	−0.0765
	$\alpha\,u_s^2$	−1.8410	−2.1124
	α	+0.2839	+0.3257
	$\alpha\,u_p$	−0.0668	−0.0766
	σ	+0.0073	+0.0084

Table 3: Fit quality on the fitted samples, on both reference areas.

Samples	n	RMSE C_L, 19.75 m^2 / 17.21 m^2	RMSE C_D, 19.75 m^2 / 17.21 m^2
$\alpha_w < 10.3°$ (below the stall)	4923	0.0264 / 0.0303	0.0031 / 0.0035
Up to $\alpha_s + 8°$	2148	0.0595 / 0.0683	0.0112 / 0.0129
All fitted	7071	0.0395 / 0.0453	0.0067 / 0.0077

Stall. $\alpha_s = 10.27°$, $w = 1.0°$. The ROM follows the operating range below the stall closely. Past it, the lift bends over instead of dropping as the data do (Fig. 8): the fit serves the range the ROM is flown in, and the upper bound on α_w of Sect. 7, not the lift curve, is what keeps it out of the stall.

Bridle pitch. $\theta_0 = -0.3398$ rad and $\theta_1 = 0.2415$ rad m^{-1} (residual 0.20° RMS): $\theta_b = 1.29°$ at 1.50 m and 16.51° at 2.60 m (Fig. 9).

Steering gain. $k_0 = 0.8466$, $k_1 = -1.1454$, $k_2 = 0.2470$, $s_0 = -0.8453$ and $s_1 = 0.6256$ (residual 0.04° RMS). The fit misses the small rise of $|k_{\phi,s}|$ with steering at the powered end, where the ROM's gain is about 8 % too strong at small steering.

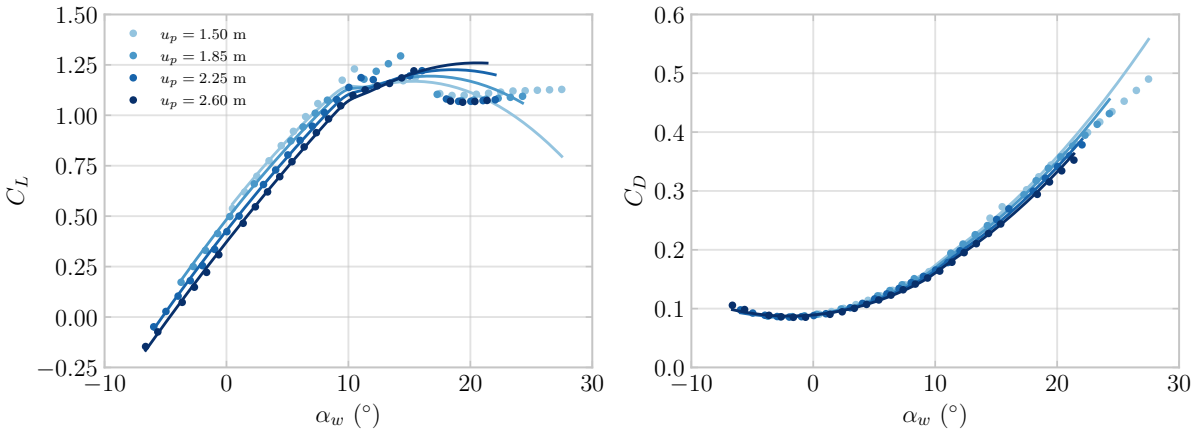


Figure 8: Lift and drag polars across the band ($u_s = 0$, 5 m s^{-1} wind): frozen-shape data on the realistic branch (markers) and the ROM (lines). Coefficients on $S = 19.75$ m^2 (multiply by 1.147 for the projected 17.21 m^2); no KCU drag, which the ROM adds separately.

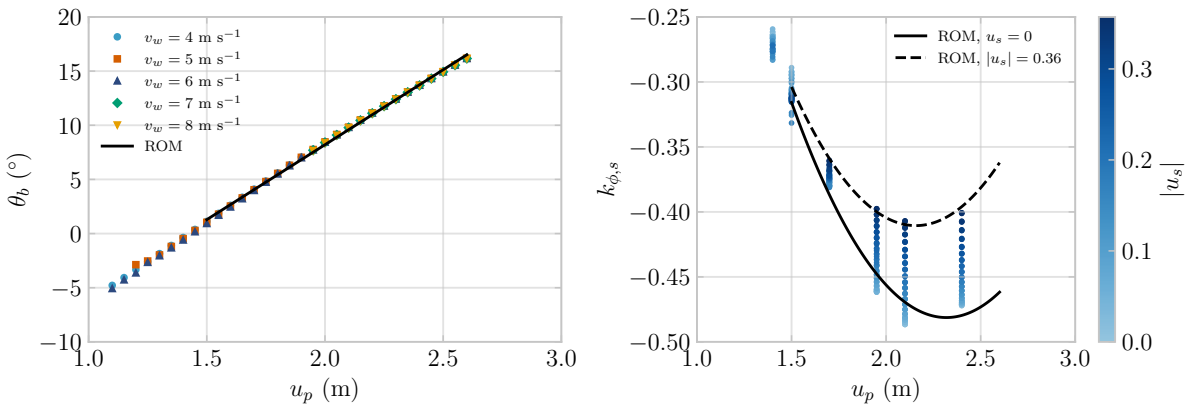


Figure 9: Left: bridle pitch against power-tape length, unsteered trims and the ROM. Right: steering gain, steered trims coloured by $|u_s|$, and the ROM's $k_{\phi,s}(u_p, u_s)$ at zero and at the largest steering.

8.2 Check against the coupled trims

A coefficient fit can look good and still trim wrongly, because θ_b, the roll gain and the polars only meet in the force balance. So each coupled trim in the band, below the stall cut, is re-solved with the ROM's own quasi-steady solver at the same condition: centre of the window, same wind, u_p, u_s, gravity off, the masses the trims carried (`validate_rom_aerostructural.py`).

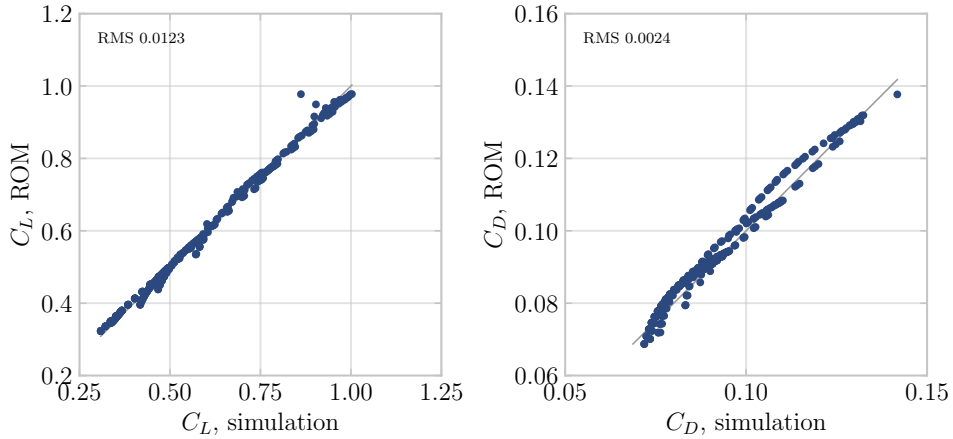


Figure 10: The ROM's coefficients against the coupled trims in the band, on $S = 19.75\,\mathrm{m}^2$ (multiply by 1.147 for $17.21\,\mathrm{m}^2$).

Table 4: ROM re-solve against the coupled trims: mean ± standard deviation of the errors; for the turn rate, the median ratio and its quartiles.

Trims	n	v_τ error (%)	F_t error (%)	α_w error (°)	$\dot{\chi}$, ROM / coupled
Unsteered	97	-4.9 ± 0.7	-5.5 ± 3.2	$+0.00 \pm 0.09$	–
Steered	278	-1.2 ± 3.3	$+1.8 \pm 7.5$	$+0.12 \pm 0.19$	1.006 (0.986–1.020)

The angle of attack and the turn rate agree closely. Speed is about 5 % low, and that comes from the tether, not from the coefficients. The check flies the ROM's rigid lumped tether, while the sweep carries a Williams tether of 10 elements, which applies its loads at interior nodes only and so carries too little drag. As the elements increase, it converges on the lumped tether: at 1.95 m and $5\,\mathrm{m\,s^{-1}}$ wind the speed goes 19.15, 18.71, 18.48 and $18.37\,\mathrm{m\,s^{-1}}$ for 10, 20, 40 and 80 elements, against $18.16\,\mathrm{m\,s^{-1}}$ lumped, with the same angle of attack. The lumped tether is therefore the reference, used in every ROM solve in this note, and the sweep's own speeds and tensions read about 5 % and 9 % high at that tape. The ROM on the same 10-element tether reproduces the unsteered trims' speed to within about 2 %.

What sets the turn rate. In a steady turn the KCU's inertia is carried through $\mathbf{F}_b$, so the wing only supplies its own centripetal force, $L\,\phi_{a,w} \approx m_w v_\tau \dot{\chi}$. The turn rate of a given gain therefore goes with the inverse of the *wing's* mass, not the kite's. The simulations carry 13.44 kg of wing and bridle. The check has to use that mass: with the 16.78 kg of the flown system file it reads a turn rate 20 % low that has nothing to do with the gain.

8.3 Flight validation

The 2019 flight is reconstructed sample by sample with the ROM's quasi-steady solver on the flown hardware. The logged steering, plus the rig's zero offset ($-0.0329/1.4$), is the input, and the ROM solves the turn rate, the tension and the speed. Cycles 60–67 are held out of the flight

correction of Sect. 9. The flown tapes are not logged in the model's coordinate, so flight depower is mapped onto u_p through the tapes that reproduce the flown tension and speed (Sect. 9): 1.95 m on reel-out and 2.48 m on reel-in.

Table 5: Flight validation, 2019 cycles 60–67, logged steering as input: medians of the relative errors, the tension's interquartile range, and the median absolute turn-rate error, over the samples solved.

ROM	Phase	Samples	Solved (%)	F_t error (%)	F_t IQR (%)	v_τ error (%)	$\lvert\Delta\dot{\chi}\rvert$ (rad s^{-1})
Simulation-only	reel-out	6804	97.1	+11.2	-11 to +38	+12.5	0.22
	reel-in	2030	94.8	+16.2	-10 to +44	+9.6	0.18
Flight-corrected	reel-out	6804	97.0	+10.0	-12 to +37	+11.7	0.18
	reel-in	2030	94.6	+16.3	-10 to +44	+9.5	0.25

Table 6: The same per held-out cycle: median tension and speed errors. The spread between cycles minutes apart is as large as the difference between the ROMs.

	Reel-out, F_t / v_τ (%)		Reel-in, F_t / v_τ (%)	
Cycle	Simulation-only	Flight-corrected	Simulation-only	Flight-corrected
60	+19 / +17	+18 / +16	+57 / +61	+57 / +62
61	−9 / +1	−10 / −0	+33 / +28	+32 / +26
62	+10 / +10	+8 / +9	+55 / +44	+55 / +46
63	+5 / +5	+4 / +4	+1 / −7	+2 / −7
64	+28 / +24	+27 / +23	+16 / +9	+15 / +9
65	+2 / +10	−0 / +9	+1 / −5	+1 / −4
66	+29 / +22	+28 / +21	+4 / −5	+4 / −5
67	+6 / +11	+5 / +11	−21 / −36	−21 / −36

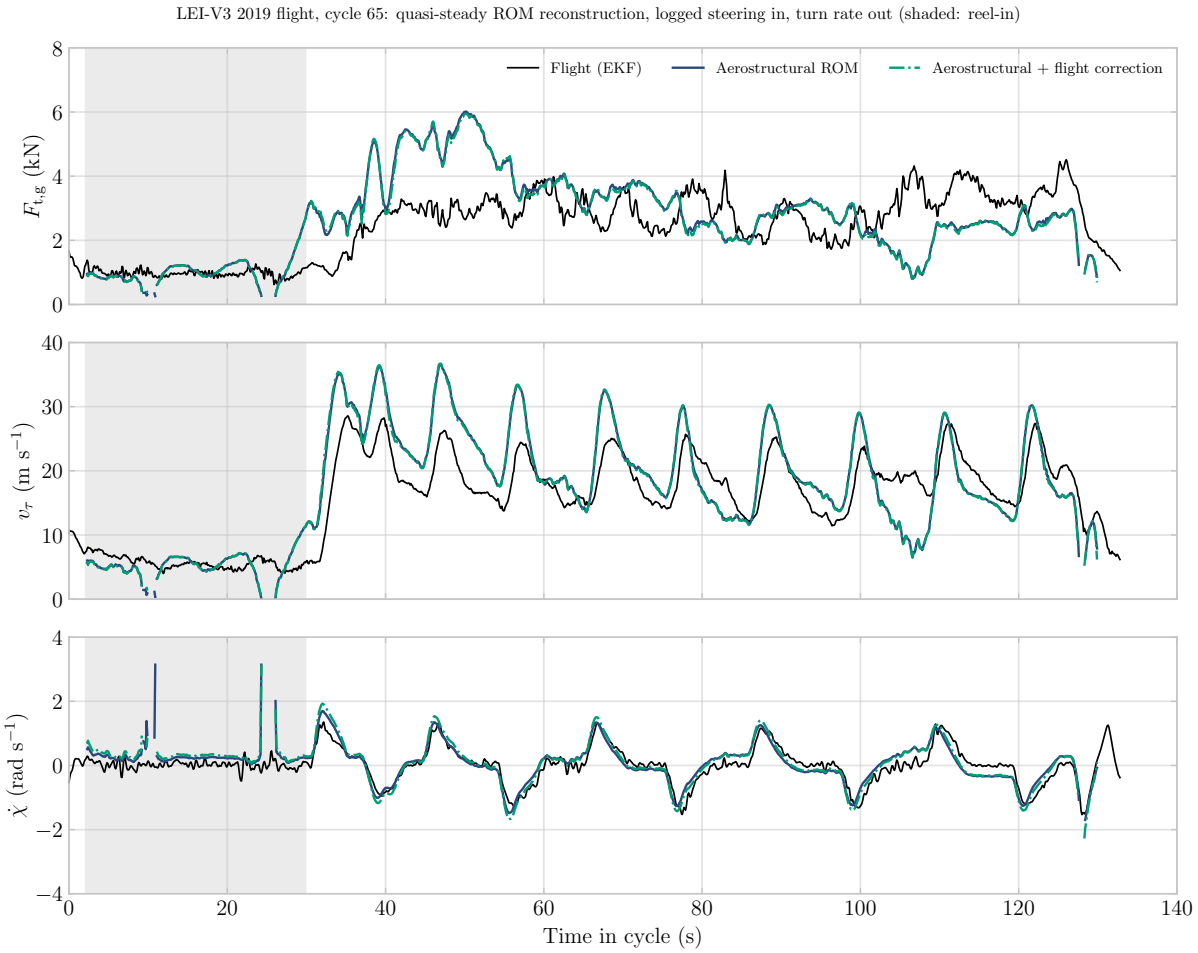


Figure 11: Cycle 65 (held out, close to the median cycle) with the logged steering as input: ground tension, tangential speed and turn rate, against the EKF reconstruction. Shaded: reel-in. Gaps are samples without a converged root.

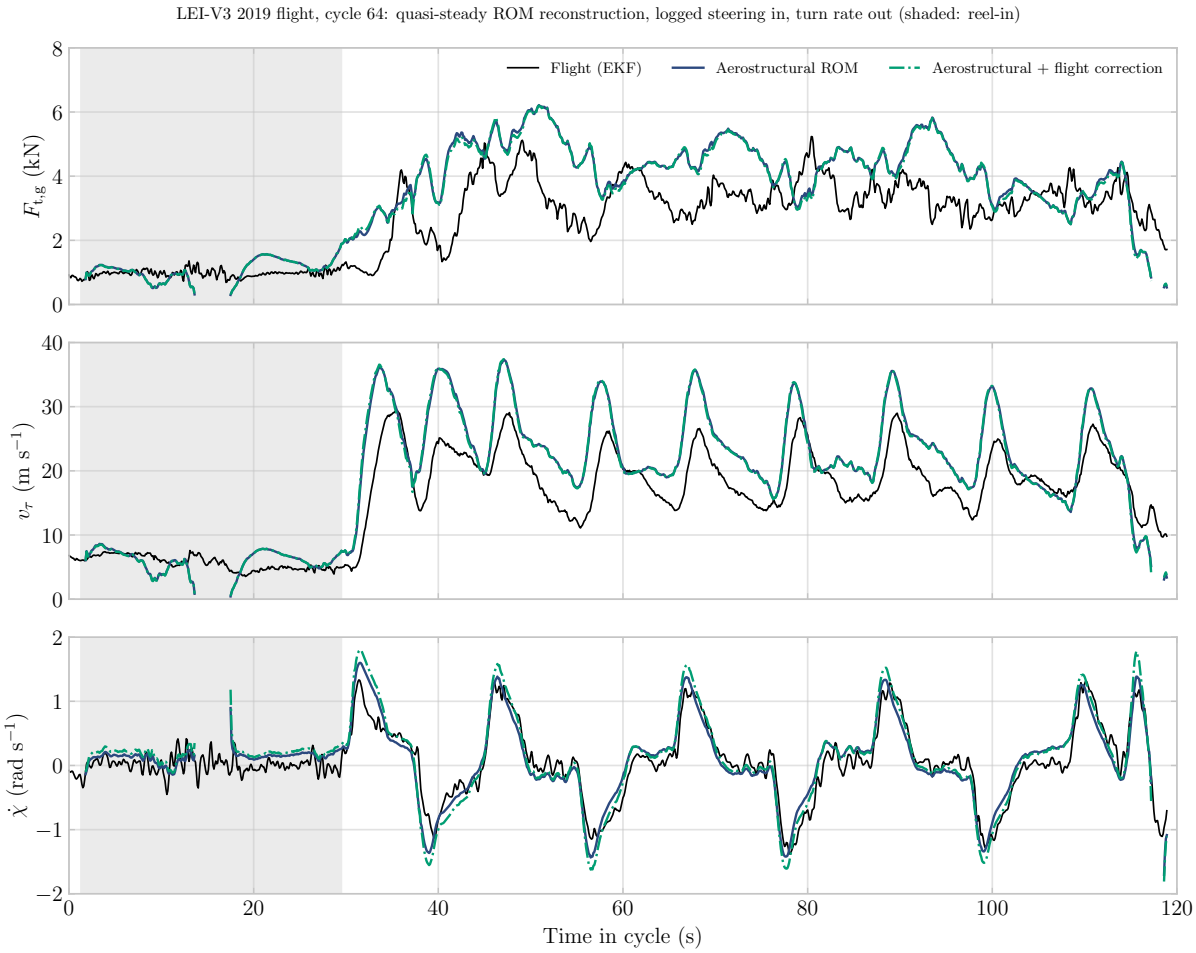


Figure 12: As Fig. 11 for cycle 64, one of the two worst held-out cycles on reel-out.

With no flight data in its coefficients, the simulation-only ROM follows the flown turn rate (Table 8) and, on the matched tapes, the flown tension and speed (Table 5). What remains is mostly scatter between cycles: the per-cycle tension and speed errors correlate strongly with the EKF's estimated wind (0.92 and 0.95 on the reel-out of cycles 40–47 and 60–67, with the old tape), as the ROM's tension responds to it about twice as strongly as the flight's over the same range. Whether that is noise in the wind estimate or something the quasi-steady model misses is open. Reel-in samples read as speed-unstable at fixed controls in almost every case ($\partial \dot{v}_\tau / \partial v_\tau > 0$, paper Eq. 36), which is why that derivative is a diagnostic here and does not select the root.

9 Flight correction

`identify_rom_flight_correction.py` adds to the simulations only what they cannot give, on the 2019 training cycles 20–27, 40–47 and 75–82, with cycles 60–67 held out. Every step scores the ROM's own quasi-steady solve on the rigid lumped tether, with the logged steering as input.

The tapes. The flown depower is not logged in the model's coordinate, so a simulation can give the kite's response to the tape but not where the flown setting sits. For each phase, the tape is the one that best zeroes both the median tension and the median speed error over the phase's samples, interpolated on a grid (Table 7). A tape where the solve keeps less than 75 % of the samples is refused, because a median over the survivors would flatter it. The result is `flight_match` in `identification.yaml`, used to read the flight for both ROMs. It changes no coefficient, so the simulation-only ROM stays free of flight data.

Table 7: The tape grid on the training cycles: median errors and the share of samples solved. Reel-out and reel-in tapes move together in one run. Selected: 1.95 m and 2.48 m.

Reel-out				Reel-in			
Tape (m)	F_t (%)	v_τ (%)	Solved (%)	Tape (m)	F_t (%)	v_τ (%)	Solved (%)
1.87	+22.9	+11.3	95	2.41	+38.0	+19.5	93
1.90	+12.8	+8.4	93	2.44	+21.8	+8.4	89
1.93	+3.3	+5.4	92	2.47	+8.6	−1.4	82
1.96	−5.1	+2.7	90	2.50	−4.7	−11.9	75
1.99	−13.1	−0.2	88	2.53	−15.5	−20.4	66
2.02	−20.3	−2.6	85	2.56	−24.0	−27.9	56

The roll gain. The flight-corrected ROM, `rom_config_aerostructural_flight_corrected.yaml`, **has not been reviewed yet and is provisional**. It is the simulation-only ROM with one scale, 1.17, on the whole roll gain, set so that the turn rate regressed on the flight's has a slope of 1 over the reel-out training samples (Table 8, Fig. 13). Lift, drag and θ_b stay simulated. As Sect. 8.2 shows, the turn rate goes with the inverse of the wing's mass, so this scale partly stands for the 13.44 kg of wing and bridle in the simulations against the 16.78 kg of the flown system file; how much is open.

Table 8: Turn-rate law $\dot{\chi} = K\, v_a u_s + c_g\, g \cos\beta \sin\chi / v_\tau$ fitted to the flight and to each ROM's turn rate (logged steering, reel-out, cycles 60–67), and each ROM's turn rate regressed on the flight's.

Source	K (rad m^{-1})	c_g	Slope against the flight
Flight (EKF)	0.172	0.47	–
Simulation-only	0.149	0.93	0.90
Flight-corrected	0.176	0.94	1.05

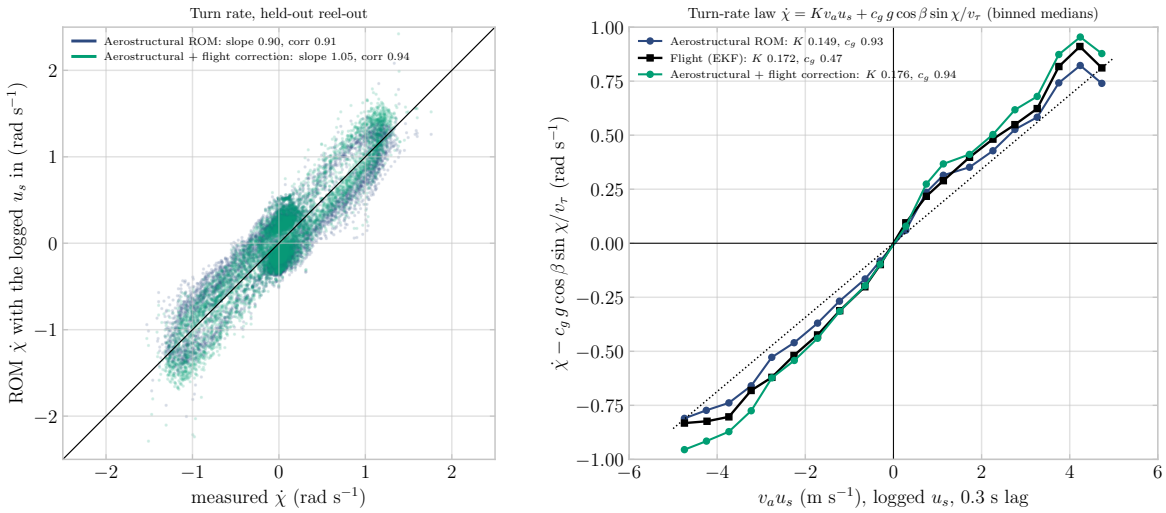


Figure 13: The turn rate of each ROM with the logged steering as input against the measured one (left, held-out reel-out samples), and the turn-rate law with the gravity term removed, binned in $v_a u_s$ (right).

Why no drag is fitted. At equal lift, the EKF's drag of wing and bridle compares to the simulated drag as in Table 9: slightly below it in straight flight, and well above it in turns, where the simulated drag falls with steering. A drag fit reproduces those numbers but does not improve the solve, for two reasons. Drag in the wing's C_D tilts the bridle resultant and raises α_w, by about $G\,\Delta C_D/C_L$ with $G = 1/(1 - \partial\epsilon/\partial\alpha_w)$ and $\epsilon = \arctan(C_D/C_L)$, some 5° at full steering,

so the kite pulls harder in the turns instead of slowing. At the bridle point it leaves the trim alone but costs too much tension: matching the steered reel-out tension takes almost none of it. The steering drag is therefore a finding, not a parameter. The EKF is also least reliable in turns, where its innovation statistics are worse and the inertial force is of the order of the drag.

Table 9: EKF drag against the simulated drag at equal lift, reel-out, training cycles, on $S = 19.75\,\mathrm{m}^2$ (multiply by 1.147 for $17.21\,\mathrm{m}^2$). KCU drag removed with the ROM's own model.

$\lvert u_s\rvert$	n	C_D, EKF	C_D, simulation	Difference
0.00–0.05	5821	0.0879	0.0987	−0.0105
0.05–0.10	2466	0.0856	0.0966	−0.0106
0.10–0.15	1159	0.0995	0.0957	+0.0039
0.15–0.20	966	0.1177	0.0935	+0.0232
0.20–0.25	1029	0.1343	0.0903	+0.0432
0.25–0.30	856	0.1524	0.0868	+0.0660
0.30–0.40	1688	0.1730	0.0806	+0.0926

What it means for the trim. That relation also gives the response to the tape, $\mathrm{d}\alpha_w/\mathrm{d}u_p \approx G\,(\partial\epsilon/\partial u_p - \theta_1)$, which matches the coupled trims within $0.5\,^\circ\,\mathrm{m}^{-1}$ RMS while θ_1 alone misses by $8.6\,^\circ\,\mathrm{m}^{-1}$: G falls from about 1.1 at 1.50 m to 0.23 at 2.60 m, because θ_b and the glide angle grow together on long tapes. So a drag change ΔC_D equals a tape change $\Delta C_D/(C_L\theta_1)$, which depends on the lift: 0.01 is worth 0.04 m of tape at 1.50 m and 0.13 m at 2.60 m. One drag offset cannot stand in for a tape error across the band.

10 Limitations and open points

- **The steering limit is not implemented yet.** The ROM's `validity` has one box, $|u_s| \le 0.357$, while at 1.50 m the last steered state that trims is 0.425 m ($|u_s| = 0.30$, at 45 % of the unsteered tension) at all three wind speeds, short of the 0.49 m flown. A limit $u_{s,\max}(u_p)$ still has to be defined and added to the optimiser.
- **Past the stall the ROM is not valid.** Its lift does not reproduce the drop, so the bound on α_w must stay active.
- **Stall data.** Several polar steps do not converge right around the stall, so the drop itself is poorly sampled, and the onset is read on steps of 1°. The stall is that of VSM on frozen deformed shapes.
- **Tether in the sweep.** The sweep's 10-element Williams tether carries too little drag (Sect. 8.2), so its absolute speeds and tensions read high. Any Williams use needs at least 40 elements.
- **Wing mass.** The simulations carry 13.44 kg of wing and bridle and the flown system file 16.78 kg. Since the turn rate depends on it, this is open for how much of a flight gain is aerodynamic.
- **Centre of the window, gravity off.** Off-centre, without weight, the wing's flow changes mainly through sideslip. The 2019 kinematics give a sideslip of $\mathbf{v}_a$ of about 12° at the median, and on the frozen trimmed shapes 10° of sideslip costs 11–20 % more C_D. The EKF's drag shows no such penalty in straight flight, so the wing probably sees far less, by yawing into the flow. This frozen estimate neither re-trims nor re-deforms; coupled solves off-centre are open. Position-dependent aerodynamics are otherwise outside the data. The gravity-driven turn of the point-mass ROM is about twice the flight's (Table 8).
- **Reel-in and the wind scatter.** The errors vary widely between cycles, following the estimated wind, and the solve loses samples quickly as the reel-in tape grows: about a fifth of them on the training cycles at the selected tape.

- **Flight data.** The EKF reconstruction is a model output too: its forces depend on its tether model and its coefficients are instantaneous at states that are not equilibria.

11 Reproduction

From the repository root:

1. `examples/3_rom_identification/run_identification_sweep.py`, then `identify_rom_aerostructural.py` and `plot_identification.py` in the same folder, for the data set, the ROM and the figures of Sects. 5–7;
2. `examples/5_flight_data/validate_quasi_steady_state_v3.py` with `--rom<rom>--steeringmeasured` on cycles 60–67, after step 3;
3. `examples/5_flight_data/identify_rom_flight_correction.py`, which runs the validator on the training cycles and writes `flight_match` and the ROM file;
4. the re-solve check, the flight plots and the tables come from `validate_rom_aerostructural.py`, `plot_rom_flight_validation.py`, `plot_rom_turn_rate_law.py` and `report_rom_aerostructural.py` (available on request), then `latexmk-pdf` in `docs/identification/`. The stall-branch figure of Sect. 5 comes from a one-off march of one frozen shape.

References

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